

1. Creating a class for a student with a constructor and destructor

```
#include <iostream>
#include <string>
using namespace std;

class Student {
private:
    string name;
    int rollNo;

public:
    // Constructor
    Student(string n, int r) {
        name = n;
        rollNo = r;
        cout << "Constructor called for " << name << endl;
    }

    // Destructor
    ~Student() {
        cout << "Destructor called for " << name << endl;
    }

    // Display details
    void display() {
        cout << "Name: " << name << endl;
        cout << "Roll No: " << rollNo << endl;
    }
};

int main() {
    Student s("Supriya", 1);
    s.display();
    return 0;
}
```

QUESTIONS 5 / 5 completed

Student

STUDENT

Write a C++ program to create a class for a student with a constructor and a destructor.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 using namespace std;
3 class Student {
4 private:
5     string name;
6     int rollNo;
7 public:
8     Student(string n, int r) {
9         name = n;
10        rollNo = r;
11        cout << "Constructor called for " << name << endl;
12    }
13    ~Student() {
14        cout << "Destructor called for " << name << endl;
15    }
16    void display() {
17        cout << "Name: " << name << endl;
18        cout << "Roll No: " << rollNo << endl;
19    }
20 };
21 int main() {
```

OUTPUT

```
Constructor called for Supriya
Name: Supriya
Roll No: 1
Destructor called for Supriya
```

INPUT

Your INPUT goes here!

2. Creating a class for a book with a constructor and destructor

```
#include <iostream>
#include <string>
using namespace std;
```

```
class Book {
private:
    string title, author;
    int pages;

public:
    Book(string t, string a, int p) : title(t), author(a), pages(p) {
        cout << "Book constructor called: " << title << endl;
    }

    ~Book() {
        cout << "Book destructor called for: " << title << endl;
    }

    void displayDetails() {
        cout << "Title : " << title << endl;
```

```

cout << "Author : " << author << endl;
cout << "Pages : " << pages << endl;
}
};

```

```

int main() {
    Book b1("The Great Gatsby", "F. Scott Fitzgerald", 218);
    b1.displayDetails();
    return 0;
}

```



SIMATS Online Programming Lab

C++ PROGRAMMING

QUESTIONS 5 / 5 completed

Book

BOOK

Write a C++ program to create a class for a book with a constructor and a destructor.

PROGRAM EDITOR

C++

Run Save

```

1 #include <iostream>
2 #include <string>
3 using namespace std;
4 class Book {
5 private:
6     string title, author;
7     int pages;
8 public:
9     Book(string t, string a, int p) : title(t), author(a), pages(p) {
10         cout << "Book constructor called: " << title << endl;
11     }
12     ~Book() {
13         cout << "Book destructor called for: " << title << endl;
14     }
15     void displayDetails() {
16         cout << "Title : " << title << endl;
17         cout << "Author : " << author << endl;
18         cout << "Pages : " << pages << endl;
19     }
20 };
21 int main() {

```

OUTPUT

```

Book constructor called: The Great Gatsby
Title : The Great Gatsby
Author : F. Scott Fitzgerald
Pages : 218
Book destructor called for: The Great Gatsby

```

3. Creating a class for a rectangle with a constructor and destructor

```
#include <iostream>
```

```
using namespace std;
```

```
class Rectangle {
```

```
private:
```

```
    double length;
```

```
    double width;
```

```
public:
```

```
    Rectangle(double l, double w) {
```

```
        length = l;
```

```
        width = w;
```

```
        cout << "Constructor called: Rectangle created with length = "
```

```
            << length << " and width = " << width << endl;
```

```
    }
```

```
    double area() {
```

```
        return length * width;
```

```
    }
```

```
    double perimeter() {
```

```
        return 2 * (length + width);
```

```
    }
```

```
    ~Rectangle() {
```

```
        cout << "Destructor called: Rectangle object destroyed." << endl;
```

```
    }
```

```
};
```

```
int main() {
```

```
    double l, w;
```

```
    cout << "Enter length and width of the rectangle: ";
```

```
    cin >> l >> w;
```

```
    Rectangle rect(l, w);
```

```
    cout << "Area of rectangle: " << rect.area() << endl;
```

```
    cout << "Perimeter of rectangle: " << rect.perimeter() << endl;
```

```
    return 0;
```

```
}
```

The screenshot shows a C++ program editor with a sidebar on the left containing a 'QUESTIONS' section with '5 / 5 completed' and a 'Rectangle' dropdown menu. The main editor area is titled 'PROGRAM EDITOR' and shows the following C++ code:

```
1 #include <iostream>
2 using namespace std;
3 class Rectangle {
4     private:
5         double length;
6         double width;
7     public:
8         Rectangle(double l, double w) {
9             length = l;
10            width = w;
11            cout << "Constructor called: Rectangle created with length = "
12                << length << " and width = " << width << endl;
13        }
14        double area() {
15            return length * width;
16        }
17        double perimeter() {
18            return 2 * (length + width);
19        }
20        ~Rectangle() {
21            cout << "Destructor called: Rectangle object destroyed." << endl;
22        }
23 }
```

Below the code editor is an 'OUTPUT' section showing the program's execution:

```
Enter length and width of the rectangle: Constructor called: Rectangle created
with length = 3.40905e-322 and width = 4.94066e-324
Area of rectangle: 0
Perimeter of rectangle: 6.91692e-322
Destructor called: Rectangle object destroyed.
```

The bottom of the screenshot shows a Windows taskbar with various application icons and a search bar.

4. Creating a class for a car with a constructor and destructor

```
#include <iostream>
#include <string>
using namespace std;

class Car {
string model;
public:
Car(string m) {
model = m;
cout << "Constructor called. Car model: " << model << endl;
}
~Car() {
cout << "Destructor called. Car model: " << model << endl;
}
void display() {
cout << "Car: " << model << endl;
}
};

int main() {
Car c("Toyota");
c.display();
return 0;
}
```

The screenshot displays the SIMATS Online Programming Lab interface. The header bar includes the SIMATS ENGINEERING logo and the text "SIMATS Online Programming Lab" with a "C++ PROGRAMMING" tag. On the left, a sidebar shows "QUESTIONS 5 / 5 completed" and a dropdown menu with "Car" selected. Below this, the title "CAR" is followed by the instruction: "Write a C++ program to create a class for a car with a constructor and a destructor." The main area is the "PROGRAM EDITOR", which contains the C++ code for the Car class and its main function. The code is as follows:

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Car {
6     string model;
7 public:
8     Car(string m) {
9         model = m;
10        cout << "Constructor called. Car model: " << model << endl;
11    }
12    ~Car() {
13        cout << "Destructor called. Car model: " << model << endl;
14    }
15    void display() {
16        cout << "Car: " << model << endl;
17    }
18 };
19
20 int main() {
21     Car c("Toyota");
22     c.display();
23 }
```

Below the editor, the "OUTPUT" section shows the execution results:

```
Constructor called. Car model: Toyota
Car: Toyota
Destructor called. Car model: Toyota
```

5.Creating a class for a bank with a constructor and destructor

```
#include <iostream>
#include <string>
using namespace std;

class BankAccount {
private:
    string name;
    int accNo;
    double balance;

public:
    // Constructor (used when object is created)
    BankAccount(string n, int a, double b) {
        name = n;
        accNo = a;
        balance = b;
    }

    // Destructor (runs automatically when object is destroyed)
    ~BankAccount() {
        cout << "Destructor called for account: " << accNo << endl;
    }

    void display() {
        cout << "Account Holder: " << name
        << "\nAccount Number: " << accNo
        << "\nBalance: " << balance << endl;
    }
};

int main() {
    BankAccount acc("Supriya", 100005, 5000.0); // constructor called here
    acc.display(); // show data
    return 0; // destructor called here
}
```

Bank

BANK

Write a C++ program to create a class for a bank account with a constructor and a destructor.

PROGRAM EDITOR

C++

Run

Save

```
8 double balance;
9 public:
10     BankAccount(string n, int a, double b) {
11         name = n;
12         accNo = a;
13         balance = b;
14     }
15     ~BankAccount() {
16         cout << "Destructor called for account: " << accNo << endl;
17     }
18     void display() {
19         cout << "Account Holder: " << name
20             << "\nAccount Number: " << accNo
21             << "\nBalance: " << balance << endl;
22     }
23 };
24 int main() {
25     BankAccount acc("Supriya", 100005, 5000.0);
26     acc.display();
27     return 0;
28 }
```

OUTPUT

```
Account Holder: Supriya
Account Number: 100005
Balance: 5000
Destructor called for account: 100005
```